

Trusted Scalar

A Qualification Calculus for Safety-Critical Reduced-Order Scalars

An independent, self-contained formulation that treats reduced-order model outputs as certificate-like quantities — with contracts, a composition algebra, invariance-preservation theorems, failure modes, and a normative spec — inside safety and forward-invariance control laws.

Reading note. This document is a specialized perspective. It is written to stand on its own: every object it relies on is defined here. It is positioned alongside a larger architecture (a reduced-order evidence manifold, a control-barrier safety filter, a temporally partitioned memory, and a fail-closed trusted computing base), but it does not depend on that architecture for its validity. Where the broader design is invoked, it is invoked only as an instantiation of the calculus, never as a premise.

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1 Scope and the central inversion

Trusted Scalar (core claim).

A reduced-order model (ROM) with validated predictors can serve as a *trusted scalar* if, and only if, it is qualified the way one qualifies a **certificate**, a **margin**, or an **invariant index**: with **bounded one-sided error**, **integrity**, and **contractual meaning** inside the control law. The scalar does not need to be accurate. It needs to be *bounded*, *conservatively biased*, and *accounted for*.

The trusted scalar is best understood as a *design primitive*, not a property of one particular estimator. A high-dimensional, possibly opaque computation (a ROM, a learned encoder, a neural surrogate) is collapsed into a single number s_t that is then permitted to participate in a safety law. The entire engineering question is: *under what conditions may a fast, fallible number be trusted to gate a high-consequence action?*

The answer rests on a single inversion of the usual objective.

The central inversion

Classical estimation minimizes *average two-sided error*. A trusted scalar instead bounds *worst-case one-sided (optimistic) error* and compiles that bound into a margin. Accuracy is replaced by *bounded conservatism*.

This inversion is what makes the rest of the calculus possible. Once we stop asking a scalar to be *right* and start asking it to be *wrong only in the safe direction, by a bounded amount*, the scalar becomes composable, monitorable, and provably safety-preserving. The remainder of the document develops that program: the contract (§2), the qualification of a ROM predictor (§3), the six-step pipeline (§4), an algebra under which trust composes (§5), the invariance-preservation argument (§6), the disqualifying failure modes (§7), a normative specification (§8), and a worked moment-based instantiation (§9).

2 The trusted-scalar contract

A scalar is *trusted* only relative to an explicit contract. We make the contract a formal object so that “trust” is a checkable predicate rather than a sentiment.

2.1 The safety order

Every trusted scalar carries a designated *conservative direction*. We encode it as a total order on the scalar’s range.

Definition 1 (Safety order). Let s take values in $\mathcal{S} \subseteq \mathbb{R}$. A *safety order* $\succeq_{\text{safe}}$ on $\mathcal{S}$ designates, for any two values, which is “at least as safe to act on.” For a *risk index* (larger means more conservative), $a \succeq_{\text{safe}} b \iff a \geq b$. For a *margin* (larger means safer), $a \succeq_{\text{safe}} b \iff a \geq b$. For a binary *witness* $w \in \{0, 1\}$ with 1 = admit, the conservative element is 0, so $a \succeq_{\text{safe}} b \iff a \leq b$. We say s' is *at least as conservative as* s when $s' \succeq_{\text{safe}} s$.

Definition 2 (Safe-monotone map). A map $g : \mathcal{S} \rightarrow \mathcal{T}$ between ordered ranges is *safe-monotone* if it preserves the safety order, $x \succeq_{\text{safe}} x' \Rightarrow g(x) \succeq_{\text{safe}} g(x')$, and its sole effect downstream is to *tighten* the admissible set (shrink feasible regions, inflate required margins, clamp actuation) —

never to loosen it. A constraint that a trusted scalar may relax is, by definition, not a place a trusted scalar may appear.

2.2 The five contract clauses

A scalar s_t is **trusted** iff it satisfies all of:

(C1) Semantic contract. s_t *means* a safety-relevant quantity (a risk index, a robust margin, an admissibility witness), not a mere correlate.

(C2) Boundedness contract. $s_t \in [s_{\min}, s_{\max}]$, well-formed: no NaN, no undefined or unreachable states, saturating at the bounds.

(C3) Conservatism contract. When wrong, s_t errs in the conservative direction of $\succeq_{\text{safe}}$: *overestimating risk / margin requirement is acceptable; underestimating is not.*

(C4) Integration contract. s_t enters the control law only through safe-monotone maps (Definition 2): tightening feasible sets, inflating margins, clamping actuation.

(C5) Liveness-of-trust contract. Qualification is not a one-time stamp; it is continuously re-earned by online conformance, and is revoked (fail-closed) on breach.

C1–C3 are *static* obligations on the scalar’s definition; C4 is a *structural* obligation on where it is allowed to appear; C5 is a *dynamic* obligation on its lifetime. A scalar that satisfies C1–C3 but is wired into a loosening mechanism (violating C4) is *mathematically valid but operationally unsafe* — a distinction the calculus is built to make visible.

3 How a ROM predictor becomes a trusted scalar

A ROM does not become trusted by being “accurate on average.” It becomes trusted because one can **upper-bound its error inside an operational envelope** and **compile that bound into the safety margin**.

We assume the standard regularity that the surrounding control law already requires.

Assumption 1 (Bounded model and encoding error). On a certified operating envelope D :

- the dynamics (f, g) are Lipschitz;
- the model deficits are bounded, $\|\Delta f\| \leq \epsilon_f$ and $\|\Delta g\| \leq \epsilon_g$;
- the encoder/representation error is bounded, $\|\Delta z\| \leq \epsilon_z$.

Under Assumption 1, a sufficient condition for safety has the form

$$\delta(\varepsilon) \geq \underbrace{D_{\text{wc}}(\epsilon_f, \epsilon_g, \epsilon_z)}_{\text{worst-case deficit}},$$

i.e. a robust margin that *dominates* the worst-case error renders the filtered action safe. This is, read correctly, a *qualification recipe*.

Qualification recipe

ROM predictors are trusted **only through the margin that dominates their worst-case error**. The ROM need not be perfect; it must be *bounded* and *accounted for*.

4 The qualification pipeline (Steps A–F)

Step A — Fix the scalar’s semantic role

Choose what the ROM emits. Three canonical roles, in increasing strength:

Role 1 — ε (risk index). A composite scalar driving conservatism, e.g.

$$\varepsilon = \text{median}(|\xi|) + \alpha \text{Var}(\text{set}) + \beta \mathbf{1}[S = 1],$$

allowed to be approximate *as long as its error is conservative* (an ε that is too high merely buys more caution).

Role 2 — δ (robust margin). A scalar that participates *directly* in the safety inequality. As a trusted scalar it must satisfy a certificate-style requirement on the whole certified envelope:

$$\delta_{\text{ROM}}(t) \geq \delta_{\text{required}}(t), \quad \forall t \text{ in the certified envelope.}$$

Role 3 — w (admissibility witness). A binary scalar $w \in \{0, 1\}$ with $w = 1$ meaning “admit this candidate action.” The conservative element is $w = 0$; the witness is fail-closed by construction (§9).

Step B — Bound ROM prediction error in the relevant norm

Establish a pointwise bound and then collapse it to a worst-case constant on D :

$$|y_{\text{true}}(x) - y_{\text{ROM}}(x)| \leq b(x) \implies \sup_{x \in D} |y_{\text{true}}(x) - y_{\text{ROM}}(x)| \leq \bar{b}.$$

The deliverable of Step B is **not high accuracy** but an **upper-bounded deviation**. Crucially, only the *optimistic* half of the error must be paid for.

Definition 3 (One-sided optimistic error). The optimistic error of s_{ROM} relative to s_{true} under $\succeq_{\text{safe}}$ is the largest amount by which the ROM is *less conservative* than the truth:

$$\bar{b}_+ \triangleq \sup_{x \in D} [s_{\text{true}}(x) \ominus s_{\text{ROM}}(x)]_+,$$

where $a \ominus b > 0$ exactly when $a \succ_{\text{safe}} b$, and $[\cdot]_+$ is the positive part. Always $\bar{b}_+ \leq \bar{b}$.

Step C — Convert the bound into a conservative scalar mapping

Define the trusted scalar as a monotone function of the bound. For a margin, $\delta(\varepsilon) \triangleq \delta_0 + k \varepsilon$ with k chosen so δ dominates the deficit terms. For a risk index, add a one-sided safety bias.

Proposition 1 (Conservative domination). Suppose the one-sided optimistic error is bounded, $\bar{b}_+ < \infty$ on D . Define

$$s_{\text{trusted}} \triangleq s_{\text{ROM}} \oplus \Delta_{\text{bias}}, \quad \Delta_{\text{bias}} \geq \bar{b}_+,$$

where $\oplus$ moves the value in the conservative direction of $\succeq_{\text{safe}}$. Then $s_{\text{trusted}}(x) \succeq_{\text{safe}} s_{\text{true}}(x)$ for all $x \in D$; equivalently, any safety inequality that holds when evaluated with s_{trusted} also holds with s_{true} .

Proof. By Definition 4, $s_{\text{true}} \ominus s_{\text{ROM}} \leq \bar{b}_+$ pointwise on D . Moving s_{ROM} conservatively by at least $\bar{b}_+$ therefore yields $s_{\text{trusted}} \succeq_{\text{safe}} s_{\text{true}}$. Because the scalar enters only through safe-monotone maps (C4), an inequality satisfied at the more-conservative value is satisfied a fortiori at the true value. $\square$

Remark 1. Proposition 1 is the quantitative heart of the inversion: the cost of trust is $\bar{b}_+$, the *one-sided* error, not the full two-sided $\bar{b}$. A predictor that is wildly inaccurate in the *conservative* direction costs nothing.

Step D — Integrity and anti-spoofing

A scalar can be mathematically valid yet untrustworthy if software or an adversary can tamper with its signal path. Qualification therefore extends to the path:

- computed in an **isolated** block (a minimal trusted computing base);
- **bounds-checked** and saturating (cannot leave $[s_{\min}, s_{\max}]$);
- **signed / hashed / bus-authenticated** end to end;
- **replicated** (e.g. majority vote) when stakes warrant.

If the scalar is corrupted, missing, or stale, it must **default to maximum risk**. We make the lifetime explicit as a guard automaton.

Definition 4 (Trust-state automaton, fail-closed). The scalar occupies one of three states with monotone-toward-safe transitions:

$$\text{TRUSTED} \xrightarrow{\text{conformance breach} / \text{integrity fault}} \text{SUSPECT} \xrightarrow{\text{persistence} / \text{hard fault}} \text{FAILED}.$$

In SUSPECT, conservatism is escalated; in FAILED, the scalar is forced to its maximum-risk value s_{maxrisk} . The only transition that *reduces* conservatism is re-qualification, which requires re-passing Steps A–C and the conformance predicate of Step E. There is no silent path back to TRUSTED.

This yields *fail-closed trust*: the absence of positive evidence is treated as the presence of risk.

Step E — Runtime conformance (trust continuously re-earned)

Even a qualified ROM becomes invalid outside its envelope. We enforce *trust only while conforming* with an online predicate K_t built from cheap monitors:

- **residual test:** $r_t = |y_t - \hat{y}_t| \leq \bar{b}$ (the certified bound);
- **drift / non-equilibrium detection:** the stiffness proxy set flags departure from the regime in which $\bar{b}$ was certified;
- **structural mode flag:** $S = \mathbf{1}[n_z \neq n_p]$ marks a dimensional / order anomaly that invalidates the ROM’s form, not merely its accuracy.

Proposition 2 (Monotone escalation under non-conformance). Let $K_t \in \{0, 1\}$ be the conformance predicate ($1 = \text{conforming}$). Define the fielded scalar

$$s_t^* = \begin{cases} s_{\text{trusted}, t}, & K_t = 1, \\ s_{\text{trusted}, t} \oplus \rho(\tau_{\text{nc}}), & K_t = 0, \end{cases}$$

where $\rho(\cdot) \geq 0$ is nondecreasing in the time-since-nonconformance τ_{nc} and $\oplus$ is conservative motion. Then $s_t^* \succeq_{\text{safe}} s_{\text{trusted},t}$ always, and $s_t^* \rightarrow s_{\text{maxrisk}}$ as τ_{nc} grows. Hence non-conformance can only shrink the trust region (raise ε , inflate δ), never enlarge it.

Proof. $\rho \geq 0$ and $\oplus$ is conservative, so $s_t^* \succeq_{\text{safe}} s_{\text{trusted},t}$ by construction; monotonicity of ρ and saturation at s_{maxrisk} give the limit. No branch reduces conservatism, so the trust region is nonincreasing along non-conformance. $\square$

Runtime qualification principle

Qualification is not a one-time stamp. It is continuously re-earned.

A certificate is valid at time t only if K_t holds; trust over a horizon is the conjunction of its instantaneous conformance.

Step F — Composition closure

A trusted scalar is rarely used alone; it is combined with others (a margin built from several indices, a witness aggregating several checks). Step F requires that *trust survive combination*. This is the subject of §5; the obligation here is simply that every combinator used be one of the closure-preserving operations proved safe there.

5 An algebra of trusted scalars

Fix a safety order $\succeq_{\text{safe}}$. Let $\mathcal{T}$ be the set of scalars qualified as trusted with respect to it. We show $\mathcal{T}$ is closed under a small set of combinators, so that trust propagates compositionally rather than having to be re-proved for every derived quantity.

Definition 5 (Conservative meet and join). For $s, s' \in \mathcal{T}$, the *conservative combination* $s \nabla s'$ returns the $\succeq_{\text{safe}}$ -larger (more conservative) of the two, and $s \bar{\wedge} s'$ the $\succeq_{\text{safe}}$ -smaller. For a risk index these are max and min; for an admit-witness (0 conservative) they are $\wedge$ and $\vee$ on $\{0, 1\}$.

Theorem 1 (Closure under safe combination). $\mathcal{T}$ is closed under each of:

- (i) conservative join ∇ (take the worst case);
- (ii) composition with any safe-monotone map g (Definition 2);
- (iii) conservative bias $s \mapsto s \oplus c$ for $c \geq 0$.

Consequently, any finite expression built from members of $\mathcal{T}$ using only ∇ , safe-monotone maps, and nonnegative bias is itself a member of $\mathcal{T}$.

Proof. We verify the clauses C1–C5 are preserved.

(i) *Join*. Boundedness (C2) is inherited since the range is unchanged. Conservatism (C3): if each s, s' dominates its truth by Proposition 1, then $s \nabla s' \succeq_{\text{safe}} s \succeq_{\text{safe}} s_{\text{true}}$ and likewise for s' , so the join dominates the joint worst case. Integration (C4) is unaffected because the join still feeds only tightening maps. Liveness (C5): the combined conformance predicate is $K_t \wedge K'_t$; a breach in either forces escalation, preserving fail-closed behavior. Semantics (C1) hold because “the more conservative of two safety-relevant quantities” is itself safety-relevant.

(ii) *Safe-monotone composition*. By Definition 2, g preserves $\succeq_{\text{safe}}$ and only tightens downstream, so $g(s) \succeq_{\text{safe}} g(s_{\text{true}})$ whenever $s \succeq_{\text{safe}} s_{\text{true}}$ (C3, C4). Boundedness follows from g mapping the bounded range into a bounded range; conformance is inherited through g ’s argument.

(iii) *Bias*. Adding $c \geq 0$ in the conservative direction strengthens C3 and preserves C2 after saturation. C1, C4, C5 are untouched.

Finite expressions: each constructor preserves membership, so by induction on the expression tree the result lies in $\mathcal{T}$. $\square$

Remark 2. The *meet* $\bar{\wedge}$ is deliberately *absent* from the closure set: taking the less-conservative of two scalars can underestimate risk and violates C3. Averaging is likewise excluded — an average of a trusted and an untrusted scalar is untrusted. The only safe way to “trust less” is re-qualification, never blending.

6 Why it works: composition with forward invariance

The trusted scalar earns its name by composing with a forward-invariance argument. We state the surrounding stack abstractly and then the preservation theorem.

Assumption 2 (Filtered safety stack). The control law comprises: (i) constraints whose invariant set $X_{\text{inv}}(\varepsilon)$ shrinks as ε grows; (ii) a control-barrier-function quadratic-program (CBF-QP) filter that projects a nominal action into the safe set defined by a barrier h with class- $\mathcal{K}$ comparison function α ; (iii) a robust margin $\delta(\varepsilon)$ entering the barrier condition; (iv) online schedules that shrink the trust radius and inflate δ when the QP nears infeasibility.

Theorem 2 (Invariance preservation under bounded scalar error). *Adopt Assumptions 1 and 2. Suppose the robust margin is a trusted scalar (qualified by Steps A–F) that dominates the worst-case deficit on D ,*

$$\delta_{\text{trusted}}(t) \geq D_{\text{wc}}(\epsilon_f, \epsilon_g, \epsilon_z) \quad \text{for all } t \text{ with the state in } D.$$

Then the CBF-QP-filtered control renders the safe set forward invariant, despite the ROM error: trajectories starting in the safe set remain in it.

Proof sketch. By Proposition 1, $\delta_{\text{trusted}} \succeq_{\text{safe}} \delta_{\text{true-required}}$ on D , so the barrier condition enforced with δ_{trusted} implies the condition that would be enforced with exact knowledge. Under Assumption 2 the CBF-QP projects every nominal action into the set where $\dot{h} \geq -\alpha(h)$ holds with the inflated margin; the Lipschitz bounds (Assumption 1) ensure the realized $\dot{h}$ deviates from the nominal by at most the dominated deficit. Hence $\dot{h} \geq -\alpha(h)$ holds for the *true* system on the boundary $h = 0$, which by the standard CBF invariance lemma yields forward invariance. Step E (Proposition 2) guarantees that if the state leaves D , δ inflates and the trust radius shrinks, so the hypothesis “state in D ” is enforced rather than assumed. $\square$

Corollary 1. *A ROM scalar is “trusted” precisely because (a) it need not be correct, (b) it must be conservative and bounded, and (c) its conservatism is what preserves invariance. This is a safety argument, not a heuristic.*

7 Failure modes: what disqualifies a scalar

Stating the disqualifiers sharpens the contract. Each is a concrete way a plausible design silently breaks one clause.

- **Accuracy worship (breaks the inversion).** Optimizing mean two-sided error instead of bounding $\bar{b}_+$. A scalar with excellent average accuracy and an unbounded optimistic tail is *not* trusted.

- **Two-sided symmetric treatment (breaks C3).** Treating over- and under-estimation alike, with no conservative bias. The fix is the one-sided bias of Proposition 1.
- **Loosening integration (breaks C4).** Wiring the scalar into a mechanism that can *relax* a constraint, enlarge a feasible set, or raise an actuation clamp. Even a perfectly bounded scalar is unsafe here.
- **Unmonitored extrapolation (breaks C5).** No conformance predicate, so the certified bound $\bar{b}$ silently ceases to hold outside D while the scalar keeps reporting confidence.
- **Bypassable path (breaks Step D).** A signal path that software or an adversary can spoof, or that does not default to max risk on loss. Trust requires fail-closed integrity.
- **Non-monotone or blended mapping (breaks Theorem 1).** Combining trusted scalars by averaging or by the conservative meet, which can underestimate the joint worst case.

8 Normative qualification specification

A ROM-derived scalar s_t is *qualified as trusted* iff it satisfies the following, which may be lifted verbatim into a requirements document. Bracketed items are the project-specific values a program must fill in.

Q1 — Bounded one-sided error. There is a certified envelope D and a constant such that $\sup_{x \in D} [s_{\text{true}} \ominus s_{\text{ROM}}]_+ \leq \bar{b}_+ = [\cdot]$. *Evidence:* bound derivation + validation set.

Q2 — Conservative mapping. $\Delta_{\text{bias}} \geq \bar{b}_+$, so the scalar never reduces conservatism when wrong (Proposition 1).

Q3 — Contract integration. The scalar appears only inside safe-monotone, tightening mechanisms (shrink feasible sets, inflate δ , tighten clamps); an explicit list of every consumer is provided, each shown to be tightening.

Q4 — Online conformance. A predicate $K_t = [\text{residual} \leq \bar{b}] \wedge [\text{no drift}] \wedge [S = 0]$ is evaluated each cycle; on $K_t = 0$ the scalar escalates per Proposition 2.

Q5 — Integrity / fail-closed. The signal path is isolated, bounds-checked, authenticated, and (if high-stakes) replicated; loss or corruption forces s_{maxrisk} via the trust-state automaton.

Q6 — Composition closure. Any scalar derived from s_t uses only the closure-preserving combinators of Theorem 1 ($\bar{\vee}$, safe-monotone maps, nonnegative bias) — never meet, averaging, or blending.

Q7 — Auditability. Each fielded value is logged with its conformance verdict and the version/signature of the producing block, so trust is reconstructable after the fact.

9 A worked instantiation: a moment-based risk index

We close with one concrete realization, presented in the calculus’s own terms. It is illustrative, not load-bearing: the theorems above do not depend on it.

9.1 Coordinates from a reduced-order evidence manifold

Let the ROM compress a high-dimensional scene into a small invariant summary — for example the leading moments $\{\mu_0, \mu_1, \mu_2, \mu_3\}$ of a population/object distribution. From these one reads three coordinates:

- ξ — a scale-free state (a canonical correlation of the summary);
- sct — a stiffness proxy (an eigenvalue of the moment Jacobian), which doubles as the drift detector of Step E;
- T^* — an urgency / driving-force proxy.

These are coordinates on a low-dimensional manifold; a hardware-friendly bilinear *witness* can test whether a candidate’s evidence lies on it within a null-band before any downstream aggregation.

9.2 The risk index as a trusted scalar

Define

$$\varepsilon = \text{median}(|\xi|) + \alpha \text{Var}(\text{sct}) + \beta \mathbf{1}[S = 1], \quad S = \mathbf{1}[n_z \neq n_p].$$

Each term is conservative-by-construction: dispersion of ξ , stiffness variance, and a structural-anomaly flag all *raise* ε (more caution) exactly when the ROM is least reliable. With the one-sided bias of Step C, $\varepsilon_{\text{trusted}} = \varepsilon + \Delta_{\text{bias}}$ satisfies $\varepsilon_{\text{trusted}} \geq \varepsilon_{\text{true}}$ on D (C3, Q2).

9.3 Realizability and passivity as the conformance predicate

The conformance predicate K_t is supplied for free by the manifold’s own validity conditions. The moment vector must correspond to a valid distribution:

$$\mu_0 \geq 0, \quad \mu_1^2 \leq \mu_0 \mu_2, \quad \det \begin{pmatrix} \mu_0 & \mu_1 \\ \mu_1 & \mu_2 \end{pmatrix} \geq 0,$$

and the induced transfer function must be passive (all poles in the left half-plane). A realizability or passivity violation means the ROM is not merely inaccurate but *ill-posed*: $K_t = 0$, and by Proposition 2 the scalar escalates rather than reports a number it cannot justify.

9.4 The margin and the witness

The robust margin is the safe-monotone image $\delta(\varepsilon) = \delta_0 + k\varepsilon$, inheriting trust by Theorem 1(ii). The admissibility witness $w \in \{0, 1\}$ is the most degenerate trusted scalar: $w = 1$ only when the candidate’s evidence lies within the null-band *and* $K_t = 1$; any fault drives $w = 0$. Because its conservative element is 0, the witness is fail-closed without any extra machinery — the limiting case of the whole calculus.

Bottom line

Yes — ROM predictors can be trusted scalars, but *only* after qualification proves: bounded *one-sided* error on a certified envelope; conservative failure direction; integration through monotone safety-tightening; composition only via closure-preserving combinators; continuous runtime conformance; and fail-closed enforcement integrity. Under those conditions the scalar’s conservatism — not its accuracy — is what preserves forward invariance, and trust propagates to every quantity built from it.

A Notation

Symbol	Meaning
$s_t, s_{\text{ROM}}, s_{\text{trusted}}$	a scalar; its raw ROM value; its qualified value
$\succeq_{\text{safe}}$	safety order (“at least as conservative as”)
$\bar{b}, \bar{b}_+$	worst-case two-sided / one-sided optimistic error on D
Δ_{bias}	conservative bias added in Step C, $\geq \bar{b}_+$
$\varepsilon, \delta(\varepsilon), w$	risk index; robust margin; admissibility witness
D	certified operating envelope
$\epsilon_f, \epsilon_g, \epsilon_z$	model and encoder error bounds (Assumption 1)
D_{wc}	worst-case deficit the margin must dominate
K_t	online conformance predicate (1 = conforming)
ξ, sct, T^*	manifold coordinates: scale-free state; stiffness; urgency
S	structural-anomaly flag $\mathbf{1}[n_z \neq n_p]$
$\bar{\vee}, \bar{\wedge}$	conservative join (allowed) / meet (forbidden)